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Phylogenies help to uncover diversity patterns in Neotropical Amphibians

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Biological diversity is a multifaceted concept. In the last decades, there is a growing interest in incorporating phylogenetic information into traditional diversity measures. Here I present results from a preliminar study about diversity patterns of a tadpole metacommunity in Southeastern Brazil. I analyzed how the phylogenetic structure of this metacommunity is influenced by pond canopy cover gradiente, pond area, floating vegetation, and depth recorded in 13 ponds. I used Principal Coordinates of Phylogenetic Structure to describe the structure of the species composition matrix wheighted by phylogeny. Then, I performed a db-RDA (Capscale) to relate environmental variables with descriptors of phylognetic structure. I found that clades are divergent in their response to two main environmental variables: canopy and floating vegetation. The db-RDA model explained 52% of the data ($P < 0.05$). Furthermore, the two main clades of hylids (Cophomantini and Dendropsophryni) were positively affected by canopy cover, contrarily to Cruciabatrachia and Microhylidae. Previous studies found that canopy cover and floating vegetation were the main drivers of tadpole metacommunity structure. Therefore, these same variables seem to be sorting not only species but also entire clades. Phylogenies are usually considered a multivariated proxy for relevant biological traits, often not directly measured on individuals. I argue for the inclusion of functional traits into the analysis of diversity patterns in amphibian studies, along with phylogenies, spatial structures, and environmental variables.